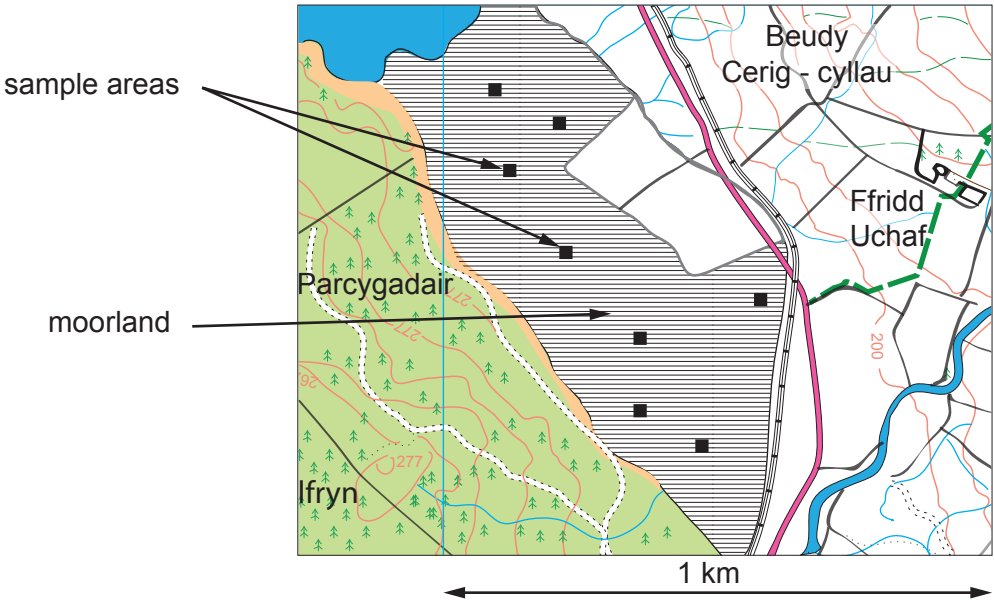


Answer **all** questions.

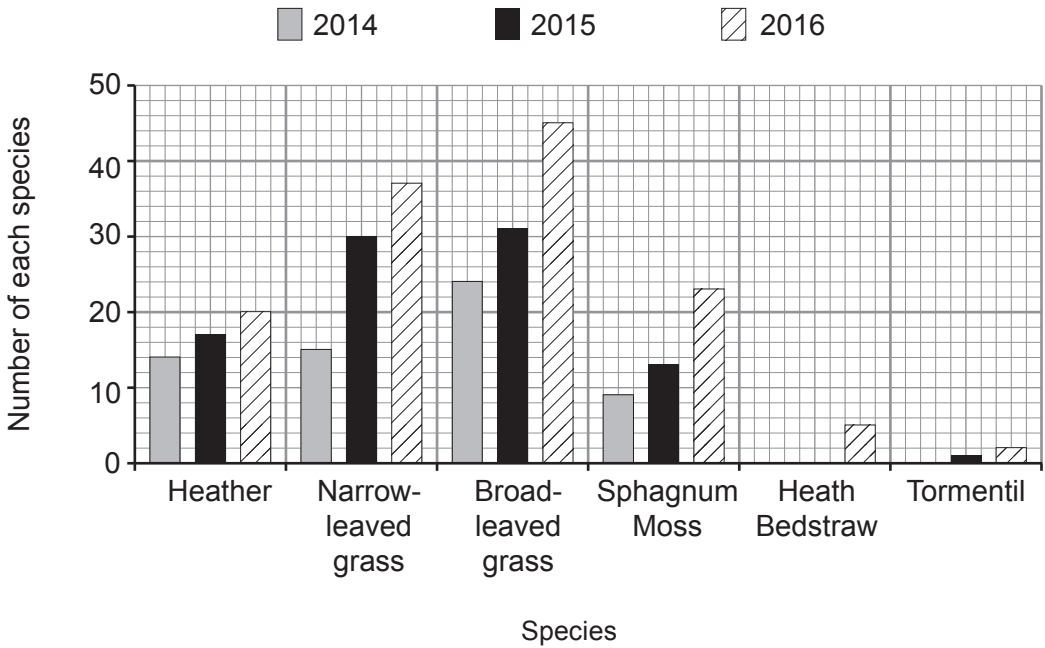
1. A group of students investigated the biodiversity of a moorland over a period of three years. Two years before the investigation started, the drainage ditches in the moorland had been closed in order to increase the water content of the land. **Image 1.1** is a map showing the moorland (shaded area) being investigated. The black squares represent locations of eight sample areas. Each sample area is 625 m² (25 m × 25 m).

Image 1.1



The results of the survey are shown in **Graph 1.2**.

Graph 1.2



Examiner
only

(a) Describe the method that the students would have used to investigate the abundance of plant species in the moorland. [4]

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(b) (i) Calculate the percentage change in narrow-leaved grass between 2014 and 2016. [2]

Percentage change = %

(ii) State **two** pieces of evidence that show that the biodiversity of the moorland increased between 2014 and 2016. [2]

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(c) (i) Suggest why the students sampled several areas over the whole moorland. [1]

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(ii) Suggest how, over the three years, the students made sure the investigation was a fair test. [1]

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